

INTERNSHIP

Weekly Report

Week No: 4

Date: 19-11-2025

Class: AIML-A

Batch No: A-16

Title of the Project: End-to-End Portfolio Management and Tax Tracking System

S.No.	Regd.No.	Name of the Student	Remarks / Suggestions (To be filled by Guide)
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Work progress :

- Began learning Graph Theory from scratch, specifically focusing on how to construct an "Adjacency Matrix" to mathematically represent the connections between stocks.
- Studied the "Graph Laplacian" operator to understand how it functions as a difference operator (like a derivative) to detect when a stock's price is out of sync with its neighbors.
- Dove into the basics of Algebraic Topology, learning about "Simplicial Complexes" and how data can have a "shape."
- Learned the concept of "Persistent Homology" and "Betti Numbers" to understand how to detect "holes" in the market structure, which serve as indicators of market fragility or crash regimes.

Signature of Guide with date